Synergy of Content and Style: Enhanced Remote Sensing Change Detection via Disentanglement and Refinement

Sijun Dong, Changxin Lu, Siming Fu, and Xiaoliang Meng

***Abstract*—Bitemporal change detection is often hindered by significant style discrepancies in images, stemming from variations in acquisition time and conditions. To mitigate this, we introduce a content–style disentanglement network (CSDNet), a novel bitemporal feature interaction network that leverages content–style disentanglement and a channel gating mechanism. In the feature encoding stage, our content–style disentanglement module (CSDM) disentangles multiscale features into content and style components using instance normalization (InstanceNorm). It then employs a dynamic gating mechanism to selectively preserve style information beneficial for change detection while suppressing background noise. A subsequent feature-level swapping strategy enhances information flow and further aligns the style representations between the bitemporal images. In the decoding stage, the contextual content refiner module (CCRM) uses a joint channel and spatial gating mechanism to attentively filter and refine the style features. These refined features are then recombined with the content features, enabling a fine-grained delineation of change regions. Extensive experiments on five public datasets—LEVIR-CD, SYSU-CD, S2Looking, WHUCD, and MSRSCD—demonstrate that CSDNet significantly surpasses various state-of-the-art methods in $F1$-score ($F1$), intersection over union (IoU), and precision (Prec). Our results validate the effectiveness of the CSDM and CCRM modules in suppressing irrelevant background interference and enhancing the representation of true changes, thereby achieving high-precision and robust change detection in complex land-cover scenarios. The source code and pretrained weights are available at https://github.com/dyzy41/CSDNet**

***Index Terms*—Channel–spatial gating, content–style disentanglement, contextual content refinement, remote sensing change detection.**

I. Introduction

REMOTE sensing change detection aims to identify and localize land-cover changes by comparing images acquired over the same area at different times. It supports a range of applications such as urban expansion monitoring, disaster assessment, and environmental surveillance. Despite recent progress, performance can be degraded by illumination variations, seasonal effects, sensor noise, and complex background structures, which often introduce pseudo changes and increase false alarms. Developing robust algorithms that generalize across diverse imaging conditions, therefore, remains an active research topic in remote sensing [1].

In recent years, deep learning-based algorithms have made significant progress in remote sensing image change detection, especially with the use of convolutional neural networks (CNNs) [2] and transformer architectures [3]. CNNs, due to their powerful local feature extraction capabilities, are widely used in change detection tasks. For example, models based on Siamese networks [4] can effectively capture common image feature details in dual-input images by extracting and comparing their features. Representative works like DASNet [5] and SNUNet [6] enhance the model's adaptability to complex scenes through multiscale feature fusion techniques such as feature pyramid networks (FPNs) and dense connections. However, CNNs have limitations in modeling long-range dependencies, leading to insufficient capability in capturing global context. The introduction of the transformer architecture, with its self-attention mechanism, has compensated for this deficiency by effectively modeling relationships between global features. For instance, BIT [7] and SwinSUNet [8] have significantly improved change detection accuracy by combining local and global features.

However, a key challenge remains: bitemporal images often contain significant style discrepancies due to varying lighting, seasons, or sensor noise. These factors introduce irrelevant feature-level perturbations, degrading detection accuracy. Our analysis, visualized via the UMAP algorithm [9] in Fig. 1 on five datasets, confirms this. We observe that low-level features in the bitemporal feature pyramid exhibit a clear domain gap. As models struggle with large domain gaps, this style-related interference severely impacts the change detection task. Consequently, a critical challenge for improving accuracy and robustness is the effective disentanglement and leveraging of content and style features.

Remote sensing image features can be broadly categorized into content features, which represent structural and semantic information (e.g., buildings and roads), and style features, which capture nonstructural differences caused by factors like lighting, seasons, and sensor noise. While content is the core target of change detection, style features can introduce interference that degrades accuracy. Recent methods, such as CCNet [10] and DHFF [11], have explored content–style separation to mitigate this issue. Unlike the above "style removal" approaches, cross-task evidence suggests the effec-

Received 2 November 2025; revised 8 January 2026; accepted 9 February 2026. Date of publication 13 February 2026; date of current version 5 March 2026. This work was supported in part by the Major Program (JD) of Hubei Province under Grant 2023BAA025 and in part by the Key Research and Development Plan of Guangxi Zhuang Autonomous Region under Grant 2023AB26007. *(Corresponding author: Xiaoliang Meng.)*

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Digital Object Identifier 10.1109/TGRS.2026.3664457

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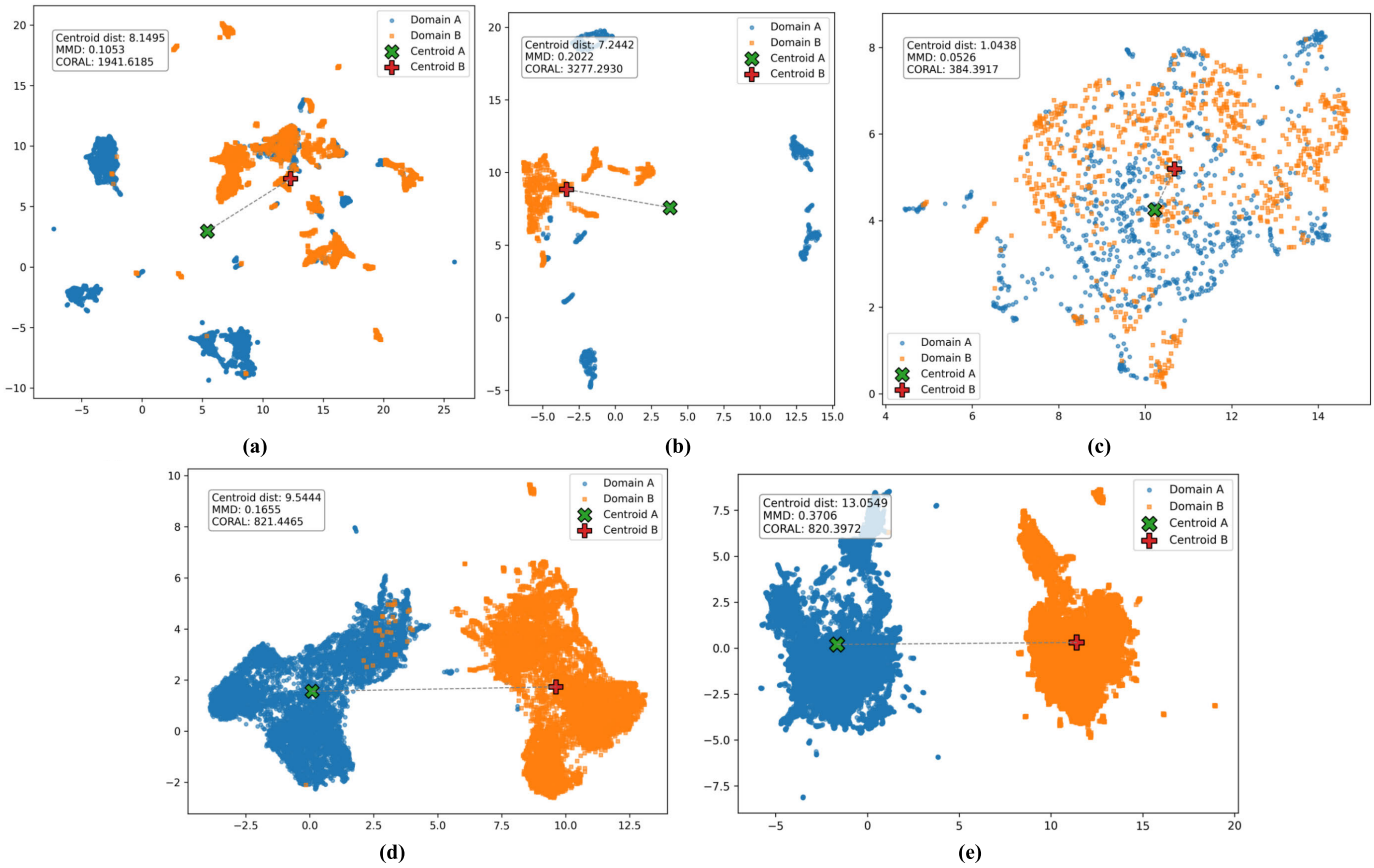


Fig. 1. Distribution of low-level features from bitemporal images after UMAP projection into two dimensions. Circles (Domain A) and squares (Domain B) represent feature samples from the bitemporal images, and the markers $\times$ and $+$ denote their respective centroids. The line connecting the centroids indicates the distance between them, reflecting the overall domain shift. Three quantitative domain discrepancy metrics are also presented: centroid distance, MMD, and CORAL distance, which characterize interdomain differences from the perspectives of geometric center, distribution consistency, and covariance structure, respectively. (a) S2Looking domain visualization. (b) LEVIR-CD domain visualization. (c) MSRSCD domain visualization. (d) WHUCD domain visualization. (e) SYSU-CD domain visualization.

tiveness of a "decouple-and-targeted-utilization" strategy: in pan-sharpening, the FD-DIFL model [12] first decomposes signals into phase and amplitude components in the frequency domain and then employs learnable high-pass filtering together with style normalization to suppress domain-specific frequency bands, thereby obtaining more domain-agnostic reconstruction representations and improving cross-domain generalization. The success of these methods demonstrates the potential of decoupling for reducing domain shifts and enhancing representation robustness. However, most existing approaches simply remove style information, potentially overlooking beneficial cues such as edges or textures. Furthermore, the challenge of effectively fusing bitemporal features to highlight changes while maintaining robustness in complex scenes remains largely unaddressed. These issues indicate that task-adaptive and synergistic designs of decoupling strategies for image processing remain underexplored.

To address these challenges, we propose content–style disentanglement network (CSDNet), a novel CNN-based change detection model that enhances accuracy and robustness through content–style separation, refinement, and feature interaction. CSDNet utilizes HRNet [13] as its backbone and a FPN [14] for multiscale feature fusion. Our key innovations are the content–style disentanglement module (CSDM), which separates and selectively preserves beneficial style features via instance normalization (InstanceNorm) and a gating mechanism, and the contextual content refiner module (CCRM). The CCRM, located in the decoding stage, further refines change features using a joint channel–spatial gating mechanism to achieve precise, fine-grained delineation of change regions. Experimental results on five public datasets demonstrate that CSDNet significantly outperforms state-of-the-art methods, validating the effectiveness of our approach in suppressing background interference and enhancing difference representation.

In this article, we propose CSDNet, a novel change detection model that integrates content–style disentanglement with channel–spatial refinement, addressing the longstanding challenge of dealing with style discrepancies in bitemporal remote sensing images. The main contributions of this work are as follows.

1) *CSDM:* We introduce the CSDM, which effectively decouples content and style features in bitemporal remote sensing images. Unlike traditional "style removal" methods, our approach selectively retains beneficial style components that enhance change detection performance, particularly in cases where texture and boundary details are crucial for accurate change delineation. This mechanism enables the model to robustly adapt to variations in lighting, season, and sensor noise, significantly improving the model's performance across complex and heterogeneous environments.

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2) *Contextual Content Refinement (CCRM):* We present the contextual content refiner module (CCRM), which refines the content–style features extracted in the encoding stage. The CCRM utilizes a joint channel–spatial gating mechanism to ensure that only relevant style features are preserved during the decoding stage, while effectively suppressing noise and irrelevant disturbances. This results in fine-grained delineation of change regions, with sharper boundaries and higher precision in detecting subtle changes, particularly in images with complex land-cover changes.
3) *Comprehensive Evaluation and Application:* Extensive experiments on five public datasets (LEVIR-CD, SYSU-CD, S2Looking, WHUCD, and MSRSCD) demonstrate that CSDNet consistently outperforms state-of-the-art methods, particularly in terms of intersection over union (IoU), $F1$-score ($F1$), and precision (Prec). The superior performance across these diverse datasets confirms the generalizability of our model in handling variations in image quality, sensor noise, and scene complexity, making it a robust solution for real-world change detection tasks in remote sensing.

II. Related Works

A. Classic Change Detection Models

Remote sensing image change detection aims to identify changes in land cover or structure by analyzing images of the same geographical area at different time points [15]. Early change detection methods mainly relied on pixel-level comparison or statistical analysis, which were suitable for multispectral or hyperspectral images and had low computational complexity but limited adaptability to complex scenes [16]. However, change detection models based on traditional machine learning algorithms suffered from insufficient adaptability to complex scenes and sensitivity to noise, making it difficult to meet the needs of high-resolution remote sensing image change detection [17].

With the rise of deep learning, CNNs and Siamese networks have significantly improved the accuracy and robustness of change detection [18]. UNet++ [19], with its nested encoding-decoding structure, enhances feature fusion and is suitable for binary change detection in high-resolution images. ECFNet [20], as a lightweight CNN, optimizes the performance of urban change detection through efficient feature extraction and comparison modules. DDNet [21], designed for synthetic aperture radar (SAR) images, combines spatial and frequency domain features to improve the robustness of change detection. The Siamese network architecture, such as ChangeFormer [22], combines a transformer encoder with a multilayer perceptron decoder to capture multiscale long-range features, significantly improving detection accuracy in complex scenes. Siamese Swin-Unet [23] further integrates the advantages of Swin transformer and U-Net, achieving collaborative extraction of global and local features through a U-shaped structure, suitable for land cover and building change detection.

In recent years, the introduction of transformer [24] and Mamba models [25] has further advanced the field of change detection. DMATNet [26] optimizes the feature fusion of bitemporal images through a hybrid attention mechanism, suitable for multispectral and high-resolution images. BIT [7] enhances change detection in complex scenes through spatiotemporal context modeling. PA-Former [27] focuses on building change detection and improving detection accuracy through prior extraction and context fusion. ACAHNet [28] adopts an asymmetric cross-attention mechanism to address the limitations of traditional methods in feature fusion. The Mamba model, as an emerging architecture, has attracted attention for its linear complexity and efficient spatiotemporal modeling capabilities. ChangeMamba [29] was the first to apply Mamba to change detection, demonstrating its potential in various tasks. The emergence of these methods indicates that with the continuous development of new architectures in computer vision, change detection technology is moving toward higher efficiency and precision.

B. Change Detection Models Based on AI Foundation Models

In recent years, with the development of AI Foundation Models, the field of change detection has gradually incorporated these models, especially algorithms based on vision-language models (VLMs) and large-scale pretrained models. These models [30], [31], through pretraining on massive datasets, can capture complex visual and semantic features, providing a powerful generalization foundation for change detection tasks. SAM-CD [32], by adapting the visual encoder of SAM and introducing task-specific modules, achieves binary change detection in high-resolution remote sensing images, achieving excellent performance on datasets such as LEVIR-CD and DSIFN-CD. AnyChange [33] utilizes SAM's semantic segmentation capabilities to achieve zero-shot change detection by calculating the semantic similarity of bitemporal images, demonstrating generalization ability without additional annotations. ChangeCLIP [34], based on the CLIP model, captures fine-grained semantic changes in bitemporal images through a differential feature compensation module and a vision-language-driven decoder, achieving state-of-the-art performance on multiple benchmark datasets. TTP [35] enhances spatiotemporal feature extraction capabilities by integrating bitemporal features and combining foundation models like SAM, suitable for high-resolution image change detection. BAN [36] proposes a new learning paradigm that reduces computational complexity while maintaining high accuracy through task prompting and a cross-attention mechanism. RSBuilding [37] unifies building extraction and change detection within a SAM-based framework, enhancing the model's generalization through a task-prompted cross-attention mechanism.

C. Feature Disentanglement in Change Detection

In recent years, feature disentanglement has emerged as a promising approach to improve change detection performance by separating content and style information. This technique allows models to focus on relevant features while mitigating the impact of style variations. Several methods have been proposed to achieve effective feature disentanglement in change detection tasks.

In the field of remote sensing change detection, some research efforts have attempted to utilize frequency information to enhance model performance. For example, Zhou et al. [38] pointed out that previous methods typically aggregate high-frequency and low-frequency components indiscriminately, leading to high-frequency details (such as textures) potentially interfering with change detection, whereas low-frequency information is more beneficial for locating changed

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regions. To address this issue, Zhou et al. [38] proposed the adaptive frequency-collaboration network, whose core idea is to perform frequency disentanglement. First, a position-specific adaptive low-pass filter was designed to extract the low-frequency components from spatial features. Subsequently, inspired by the principle of wavelet reconstruction, the corresponding high-frequency components are obtained by subtracting the low-frequency part from the original spatial features. The disentangled frequency components are then assigned to different tasks to avoid interference. In the change captioning task, distinguishing between real changes and pseudo-changes caused by viewpoint shifts is a key challenge. To tackle this problem, Tu et al. [39] proposed a viewpoint-adaptive representation disentanglement network. The core idea of this method is to utilize feature disentanglement to explicitly capture change features. Specifically, they designed an "unchanged representation disentanglement" module, which infers a stable change feature representation by identifying and disentangling the common attributes or unchanged features in the image pair. This strategy of disentangling the "unchanged" representations from the image pair aims to learn more reliable change information to resist feature interference caused by viewpoint variations. Furthermore, addressing the problem that existing change detection models tend to cause blurred semantic boundaries when extracting multiscale hybrid features, Wang et al. [40] proposed a deep network based on feature disentanglement, DSFDcd. The core idea of this method is to explicitly decompose the hybrid features into change features and invariant features. The DSFDcd, based on decoupled representation, can project hybrid features into more clearly defined changeable and invariant feature spaces, thereby achieving more robust change detection results. It is evident that these feature disentanglement-based change detection methods, by decomposing image features and processing the decomposed features separately, effectively enhance the model's performance in complex scenarios, demonstrating the great potential of feature disentanglement in remote sensing change detection.

D. Change Detection Models Combining Style and Content Features

In the research area of feature disentanglement, content–style separation has likewise drawn broad attention, particularly for remote sensing image change detection. Bitemporal image data often exhibits significant domain shift, typically manifested as differences in image style. These style differences often introduce irrelevant disturbances at the feature level, thereby affecting the accuracy of the change detection task. Therefore, how to effectively separate and utilize content and style features has become key to enhancing change detection accuracy and robustness. In recent years, change detection algorithms based on the separation of style and content features have made significant progress in the remote sensing field. CCNet [10] decomposes remote sensing images into a shared content space and private style spaces through separated representation learning, using a multiresolution parallel structure to extract semantic and spatial details, suitable for high-resolution image change detection. DHFF [11], targeting heterogeneous optical and SAR images, uses style transfer technology to achieve feature homogenization, preserving content features through an iterative style transfer strategy, significantly improving the accuracy of heterogeneous change detection. DTCDN [41] utilizes a generative adversarial network (GAN) for deep translation, mapping optical and SAR images into a unified feature space, and combines an improved UNet++ network to enhance change detection performance, especially suitable for multimodal data scenarios. Furthermore, semantic separation representation learning methods enhance the change detection capability for specific targets like buildings by separating the representations of different semantic regions through self-supervised learning and semantic masks, suitable for scenarios with insufficient data [42]. CiDL [43] proposes a content-invariant dual learning framework that achieves cross-domain translation through two Y-shaped networks, preserving content features while homogenizing style features, suitable for both supervised and unsupervised change detection tasks. DADR-HCD [44], through a deep domain adaptation and separated representation network, explicitly decomposes domain-invariant content features and domain-specific style features, solving the domain gap problem of heterogeneous images and performing exceptionally well in unsupervised heterogeneous change detection.

These methods demonstrate the potential of separating style and content features in remote sensing change detection, especially when dealing with lighting variations, seasonal differences, and heterogeneous images. However, most existing works tend to remove style features to reduce interference, paying less attention to the potentially beneficial information within style features for change detection, such as edge or texture changes. Accordingly, compared with prior methods, our proposed CSDM module performs effective content–style disentanglement and introduces a channel–spatial gating mechanism that further optimizes the style component via dynamic gating. This allows the model to adaptively retain style cues beneficial for change detection while suppressing irrelevant interference, thereby improving detection accuracy and robustness.

III. Methodology

A. Overall Network Architecture

As shown in Fig. 2, the overall structure of the model is a Siamese architecture. The bitemporal images first pass through the feature extraction network, the HRNet model, to extract a multiscale feature pyramid. Subsequently, the feature pyramid undergoes content–style decomposition through the CSDM to obtain content features and style residuals. Simultaneously, a gating mechanism is used to filter the style residuals, retaining style information that is useful for change detection, as shown in the left part of Fig. 2. The processed style features are then recombined with the original content features to obtain purified features. CSDM is applied to each pyramid level to reduce style-induced perturbations and produce purified features for subsequent cross-temporal interaction.

Next, we use a feature-level swapping mechanism to perform cross-layer exchange on the bitemporal image feature pyramids, as proposed in EfficientCD [58]. Specifically, let the feature pyramids from time phase t_1 be $(F_1^1, F_2^1, F_3^1, F_4^1)$ and those from time phase t_2 be $(F_1^2, F_2^2, F_3^2, F_4^2)$, where the subscript denotes the pyramid level from shallow to deep. In the exchange operation, we alternately swap features at every other level (e.g., $F_2^1 \leftrightarrow F_2^2$ and $F_4^1 \leftrightarrow F_4^2$), resulting in hybrid pyramids such as $(F_1^1, F_2^2, F_3^1, F_4^2)$ and $(F_1^2, F_2^1, F_3^2, F_4^1)$. This encourages cross-temporal feature alignment by allowing

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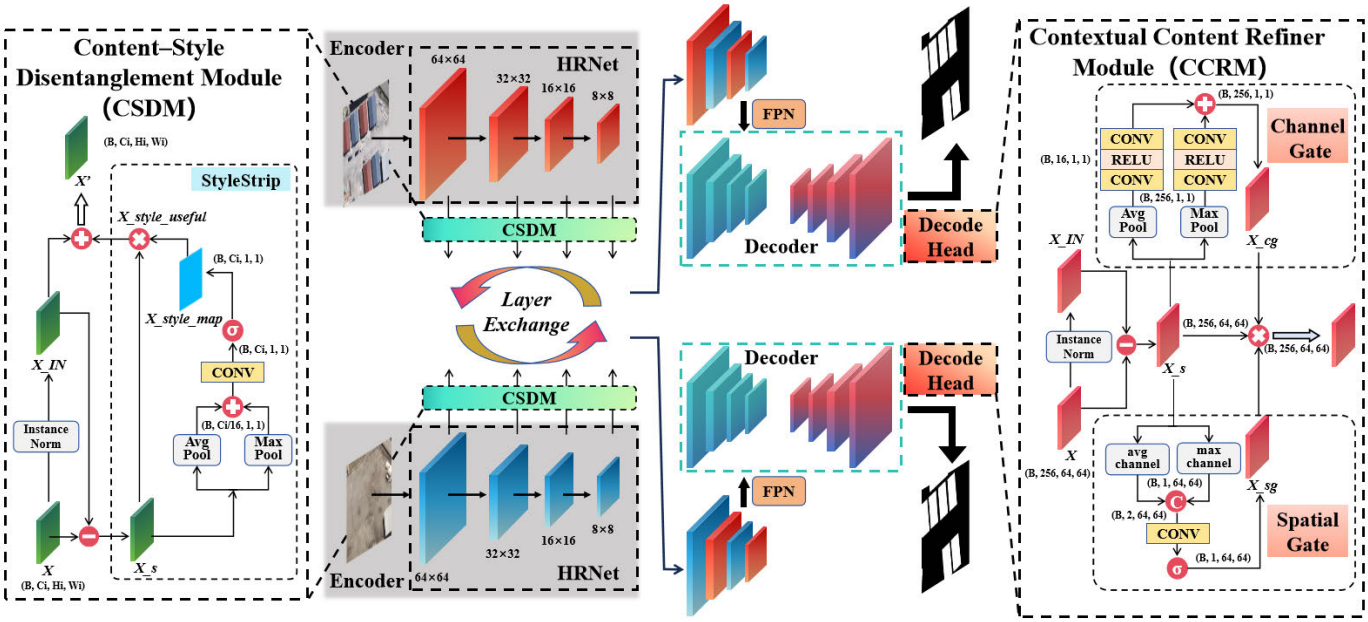


Fig. 2. Overall architecture of the proposed CSDNet model for change detection. Overview of CSDNet. A Siamese HRNet extracts multiscale features for two times; CSDM performs content–style disentanglement with gated selection, followed by cross-temporal feature swapping. FPN fuses the pyramids, and a decoder upsamples; CCRM applies joint channel–spatial refinement before producing the final change map. All modules are weight-shared across the two branches.

information from one time phase to be directly injected into nonadjacent levels of the other phase, thereby enhancing the model's sensitivity to structural changes while preserving the semantic hierarchy.

Subsequently, based on the feature-swapped bitemporal feature pyramids, we use an FPN structure for feature fusion to obtain multiscale bitemporal feature maps. In the decoder, we use conventional layer-by-layer upsampling for decoding. Finally, for the decode head, we designed a contextual content refiner module (CCRM) to further refine the decoded features output from the decoder, in order to enhance the task-relevant signals for change detection, as shown in the right part of Fig. 2. The refined features are then mapped to a change probability map. In the entire CSDNet, modules such as the encoder, CSDM, FPN, decoder, and decode head all employ a parameter-sharing strategy, thereby achieving effective parameter reuse and reducing the model's parameter count.

B. CSDM

In change detection tasks, significant temporal gaps between the acquisition of bitemporal remote sensing images introduce "style" variations—such as atmospheric conditions, lighting changes, and sensor noise—that can adversely affect Siamese neural network learning. These factors introduce irrelevant disturbances at the feature level, thus affecting the accuracy of the change detection task. To address this, we designed the CSDM, as shown in the left part of Fig. 2, which selectively retains style information beneficial to downstream tasks through "content–style" decomposition and a gating mechanism, while stripping away redundant parts. For the multiscale feature pyramids of the bitemporal images, we use the CSDM module to process the features at each scale. The CSDM module can explicitly separate content information from style information and adaptively select and retain useful style features for change detection through a gating mechanism. This allows the change detection model to focus more on extracting change information, thereby optimizing the change detection task. The CSDM calculation process is as follows.

First, InstanceNorm is applied to the input feature $F \in \mathbb{R}^{C\times H\times W}$ to obtain the content feature

$$F_{\text{content}} = \text{IN}(F) \tag{1}$$

where the InstanceNorm operation is defined as

$$\mu_c = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W} F_{cij} \tag{2}$$

$$\sigma_c = \sqrt{\frac{1}{HW}\sum\nolimits_{i,j}\left(F_{cij}-\mu_c\right)^2 + \epsilon} \tag{3}$$

$$\text{IN}(F)_{cij} = \frac{F_{cij}-\mu_c}{\sigma_c} \tag{4}$$

which normalizes each channel of each sample to have zero mean and unit variance along the spatial dimensions, effectively removing style-related global statistical information such as brightness, contrast, and noise differences, while preserving spatial structure and local texture details. Therefore, IN(F) can be regarded as an effective approximation of the image's "content" information. Consequently, the style residual of the input image feature can be obtained through residual calculation

$$F_{\text{style}} = F - F_{\text{content}}. \tag{5}$$

In the GateBlock, the style residual $F_{\text{style}} \in \mathbb{R}^{C\times H\times W}$ is first aggregated through global average pooling (AvgPool) and global max pooling (MaxPool) branches, respectively,

$$A = \text{Conv}_{\text{avg}}\left(\text{AvgPool}\left(F_{\text{style}}\right)\right) \tag{6}$$

$$\tilde{A} = \text{ReLU}\left(\text{LN}\left(A\right)\right) \tag{7}$$

$$M = \text{Conv}_{\text{max}}\left(\text{MaxPool}\left(F_{\text{style}}\right)\right) \tag{8}$$

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$$\tilde{M} = \mathrm{ReLU}\left(\mathrm{LN}\left(M\right)\right) \tag{9}$$

where $\mathrm{Conv}_{\mathrm{avg}}$ and $\mathrm{Conv}_{\mathrm{max}}$ are both 1×1 convolutions for channel compression, LN represents LayerNorm calculation, and ReLU is the activation function. Subsequently, the features from both paths are added together and passed through a second 1×1 convolution to generate a channel gate map

$$S = \tilde{A} + \tilde{M} \tag{10}$$

$$G = \sigma\left(\mathrm{Conv}_2\left(S\right)\right), \quad G \in [0,1]^{C\times 1\times 1} \tag{11}$$

$$F'_{\mathrm{style}} = F_{\mathrm{style}} \otimes G \tag{12}$$

where Conv_2 is a 1×1 convolution that expands the channels back to C dimensions, σ is the Sigmoid activation function, and $\otimes$ denotes element-wise multiplication. As seen from the above, the CSDM module uses a gating mechanism to weight the style residual channel-wise, obtaining refined style features. Finally, the stripped content features are recombined with the filtered style features to get the purified feature

$$\tilde{F} = F_{\mathrm{content}} + F'_{\mathrm{style}}. \tag{13}$$

Through the calculation method described above, CSDM enables the network to remove irrelevant interference while retaining important style cues via explicit stripping and a learnable gate. This allows the network to autonomously learn to what extent to preserve "style" components, thereby balancing denoising and information integrity. As a result, the change detection model gains stronger adaptability to style differences in bitemporal images.

C. Contextual Content Refiner Module (CCRM)

In the decoder, we introduce the contextual content refiner module (CCRM) to the features $P \in \mathbb{R}^{C\times H\times W}$ after they have been upsampled and fused layer-by-layer, in order to further filter and enhance features relevant to the change detection task. The CCRM consists of four steps: content–style decomposition, channel gating, spatial gating, and finally recombining the refined style back into the content.

1) Content–Style Decomposition: First, the feature map P is converted into a content feature C using InstanceNorm, and the style feature S is extracted in a residual form

$$C = \mathrm{IN}\left(P\right) \tag{14}$$

$$S = P - C \tag{15}$$

where $\mathrm{IN}(\cdot)$ de-means and divides by the variance for each channel of each sample spatially, removing global statistics (like brightness, contrast, and noise differences) and retaining only texture and structural information. This makes C more stable and reliable, while S contains the remaining "style" components, including noise and changes that are detrimental to change detection.

2) Channel Gating: To adaptively select useful style components in the channel dimension, CCRM employs parallel processing with global AvgPool and global MaxPool, followed by a shared transformation to generate a channel gate map $G_{\mathrm{chan}} \in [0,1]^{C\times 1\times 1}$. The specific steps are as follows:

$$U_{\mathrm{avg}} = \mathrm{AvgPool}\left(S\right), \quad U_{\mathrm{avg}} \in \mathbb{R}^{C\times 1\times 1} \tag{16}$$

$$U_{\mathrm{max}} = \mathrm{MaxPool}\left(S\right), \quad U_{\mathrm{max}} \in \mathbb{R}^{C\times 1\times 1} \tag{17}$$

$$A = W_2\left(\mathrm{ReLU}\left(W_1\left(U_{\mathrm{avg}}\right)\right)\right) \tag{18}$$

$$M = W_2\left(\mathrm{ReLU}\left(W_1\left(U_{\mathrm{max}}\right)\right)\right) \tag{19}$$

$$G_{\mathrm{chan}} = \sigma\left(A + M\right) \tag{20}$$

where W_1 represents a channel-compressing 1×1 convolution, reducing the number of channels from C to C/r; after ReLU activation, W_2 uses another 1×1 convolution to restore the channel count to C. σ is the Sigmoid function, and each element of G_{chan} is in $[0,1]$, representing the importance weight of the corresponding channel.

Channel gating can adaptively suppress or enhance style information of specific channels based on global statistics, making the model focus on channel features that contribute to change detection.

3) Spatial Gating: In the spatial dimension, we first perform AvgPool and MaxPool on the style feature S along the channel dimension, then concatenate them and pass them through a large-receptive-field convolution to generate a spatial gate map $G_{\mathrm{spat}} \in [0,1]^{1\times H\times W}$

$$M_{\mathrm{mean}} = \frac{1}{C}\sum_{c=1}^{C} S_c, \quad M_{\mathrm{mean}} \in \mathbb{R}^{1\times H\times W} \tag{21}$$

$$M_{\mathrm{max}} = \max_{c=1,\ldots,C} S_c, \quad M_{\mathrm{max}} \in \mathbb{R}^{1\times H\times W} \tag{22}$$

$$G_{\mathrm{spat}} = \sigma\left(\mathrm{Conv}_{7\times 7}\left(\left[M_{\mathrm{mean}};\, M_{\mathrm{max}}\right]\right)\right). \tag{23}$$

M_{mean} and M_{max} capture the average and strongest responses across channels, respectively. The concatenated two-channel input is fed into a 7×7 convolution to obtain rich spatial context.

4) Style Recombination and Output: Finally, both channel and spatial gates are applied to S, and the result is added back to the content feature C to obtain the refined feature $\hat{P}$

$$S' = S \otimes G_{\mathrm{chan}} \otimes G_{\mathrm{spat}} \tag{24}$$

$$\hat{P} = C + S'. \tag{25}$$

Here, $\hat{P}$ is the refined feature map, containing style information filtered through the gates. In this way, CCRM can adaptively filter style information in both channel and spatial dimensions, suppressing irrelevant interference and preserving features that are helpful for change detection, thereby significantly improving the semantic accuracy and robustness of the decoder output.

D. Loss Function in CSDNet

In the CSDNet model, we use a simple loss function based on cross-entropy to train the model. As shown in Fig. 2, we have two equivalent decode heads. The model employs a dual-branch binary classification scheme with cross-entropy supervision. Let the raw logits from the two branches be $o^{(1)}$ and $o^{(2)}$, and the ground-truth label be $y \in \{0,1\}$. The cross-entropy loss for a single branch is defined as

$$\mathcal{L}_{\mathrm{CE}}\left(o,y\right) = -\log\left(\frac{\exp\left(o_y\right)}{\sum_{k=0}^{1}\exp\left(o_k\right)}\right) \tag{26}$$

where o_k denotes the logit corresponding to class k. During training, the final loss is the average of the two branch-level cross-entropy losses

$$\mathcal{L}_{\mathrm{train}} = \frac{1}{2}\left(\mathcal{L}_{\mathrm{CE}}\left(o^{(1)},y\right) + \mathcal{L}_{\mathrm{CE}}\left(o^{(2)},y\right)\right). \tag{27}$$

This formulation leverages the redundancy and complementarity between the two branches to improve robustness, while

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averaging prevents any single branch from dominating the gradient. At inference time, the two branch logits are averaged to produce the final prediction

$$\hat{o} = \frac{1}{2}\left(o^{(1)} + o^{(2)}\right) \tag{28}$$

which yields a smoother and more stable classification output compared to using a single branch.

IV. Experimental Results

A. Datasets

In this study, we utilized five submeter resolution datasets, WHUCD [66], SYSU-CD [67], LEVIR-CD [45], MSRSCD [68], and S2Looking [69], to demonstrate the robustness and versatility of our change detection algorithm across different environments and scenarios. The SYSU-CD dataset comprises a total of 20 000 images with dimensions of 256 × 256, in which changes of multiple targets are annotated. The SYSU-CD dataset is divided into three sets: the training set, validation set, and test set, consisting of 12 000, 4000, and 4000 image pairs, respectively. The LEVIR-CD dataset is an important resource for building change detection. It is divided into 445 training image pairs, 64 validation image pairs, and 128 testing image pairs. The MSRSCD dataset comprises 842 pairs of remote sensing images captured in southern Chinese cities from 2019 to 2023. The dataset is divided into training, validation, and test sets in a ratio of 7:1:2. The S2Looking dataset comprises 5000 image pairs with over 65 920 annotated changes derived from rural satellite images (0.5–0.8 m/pixel). It is divided into training, evaluation, and testing sets in a 7:1:2 ratio. LEVIR-CD, S2Looking, and MSRSCD images have a resolution of 1024 × 1024. During training, we perform random cropping to 256 × 256 patches, while during testing, a sliding window approach is adopted to obtain the results.

The WHUCD dataset consists of two aerial images with a resolution of 32 507 × 15 354 pixels and a spatial resolution of 0.3 m per pixel. Similar to the LEVIR-CD dataset, the WHUCD dataset is specifically designed for building change detection. There is no standardized data split method for the WHUCD dataset. To ensure fair comparison and reproducibility, we adopted a consistent data partition scheme and replicated several existing algorithms for comparison against our proposed CSDNet model. Following the works in [7], [47], and [66], we divided the original images into nonoverlapping chips of 256×256 pixels. These chips were randomly split into three subsets: 6096 chips for training, 762 chips for validation, and 762 chips for testing, following the same splitting strategy as in previous works.

B. Implementation Details

In our experiments, we implement CSDNet in PyTorch with PyTorch Lightning. Models are trained on NVIDIA RTX 4090 GPUs using AdamW with an initial learning rate of $3e^{-4}$. We select the checkpoint with the best validation IoU on the change class and report results on the test set using this checkpoint.

C. Evaluation Metrics

We used these metrics to evaluate our model, like precision (Prec), recall (Rec), overall accuracy (OA), $F1$, and IoU. Its calculation formula is as follows:

$$\text{IoU} = \frac{TP}{TP + FN + FP} \tag{29}$$

TABLE I
Comparison of Different Methods on the SYSU-CD Dataset

Method	Year	OA	IoU	F1	Rec	Prec
STANet [45]	2020	88.24	57.22	72.79	66.71	80.08
BIT [7]	2022	90.64	66.03	79.54	77.13	82.10
L-UNet [46]	2021	90.58	66.15	79.63	78.08	81.24
P2V [47]	2023	90.49	66.29	79.73	79.29	80.17
HCGMNet [48]	2023	91.12	66.33	79.76	74.15	86.28
CGNet [49]	2023	91.19	66.55	79.92	74.37	86.37
DARNet [50]	2023	91.26	68.10	81.03	79.11	83.04
CDNeXt [51]	2024	91.99	68.57	81.36	74.10	90.20
STDF-CD [52]	2025	91.42	68.83	81.53	80.35	82.76
BASNet [53]	2024	91.80	69.71	82.15	80.01	84.41
HATNet [54]	2024	90.92	67.00	80.24	78.23	82.36
AdaptVFMs-RSCD [55]	2025	90.74	–	81.14	82.50	79.83
ChangeCLIP [34]	2024	92.08	70.53	82.72	80.41	85.16
ISDANet [56]	2025	91.83	70.90	82.97	83.38	82.06
SGSLN [57]	2023	–	71.05	83.07	81.45	84.76
ChangeMamba [29]	2024	92.30	71.10	83.11	80.31	86.11
CSDNet	2025	92.39	71.16	83.15	79.60	87.03

TABLE II
Comparison of Different Methods on the S2Looking Dataset

Method	Year	OA	IoU	F1	Rec	Prec
HATNet [54]	2024	–	47.08	64.02	60.90	67.48
FHD [59]	2022	–	47.33	64.25	56.71	74.09
CGNet [49]	2023	–	47.41	64.33	59.38	70.18
BIT [7]	2022	99.24	47.94	64.81	58.15	73.20
SAM-CD [32]	2024	–	48.29	65.13	58.92	72.80
VMMCD [60]	2025	–	48.32	65.16	64.86	65.45
DMINet [61]	2023	99.20	48.33	65.16	62.13	68.51
PCAANet [62]	2023	99.22	48.54	65.36	61.54	69.68
HFIFNet [63]	2025	99.22	48.54	65.35	61.04	70.33
CDNeXt [51]	2024	–	50.05	66.71	63.08	70.78
Changer [64]	2022	–	50.47	67.08	62.04	73.01
CSDNet	2025	99.24	50.72	67.31	64.34	70.55

$$\text{Prec} = \frac{TP}{TP + FP} \tag{30}$$

$$\text{Rec} = \frac{TP}{TP + FN} \tag{31}$$

$$F1 = \frac{2 \cdot Prec \cdot Rec}{Prec + Rec} \tag{32}$$

$$\text{OA} = \frac{TP + TN}{TP + TN + FN + FP}. \tag{33}$$

D. Compared Methods

As shown in Tables I–V, we compare CSDNet with representative recent RSCD methods covering transformers, attention-based fusion, difference modeling, Mamba-style architectures, and foundation-model adaptations. Table VI summarizes these baselines and their key research focuses.

E. Quantitative Analysis and Visualization

As shown in Tables I–V, we evaluated CSDNet on five widely used remote sensing change detection datasets and compared its performance against representative state-of-the-art methods. Overall, CSDNet achieves consistently competitive or superior results, with clear advantages on SYSU-CD, S2Looking, WHUCD, and MSRSCD, particularly in terms of IoU and $F1$. Crucially, these performance gains are not uniform but are directly linked to the specific challenges

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TABLE III
COMPARISON OF DIFFERENT METHODS ON THE WHUCD DATASET

Method	Year	OA	IoU	F1	Rec	Prec
ISDANet [56]	2025	99.05	82.26	90.27	94.84	86.12
DARNet [50]	2022	99.19	83.92	91.26	91.71	90.82
P2V [47]	2023	99.31	85.91	92.42	90.93	93.97
SGSLN [57]	2023	99.38	87.47	93.32	92.91	93.72
BIT [7]	2022	99.43	88.22	93.74	92.00	95.56
HCGMNet [48]	2023	99.52	90.10	94.79	95.31	94.27
ChangeCLIP [34]	2024	99.52	90.15	94.82	94.02	95.63
CDNeXt [51]	2024	99.54	90.35	94.93	92.56	97.43
CGNet [49]	2023	99.54	90.41	94.96	94.61	95.32
EfficientCD [58]	2024	99.55	90.71	95.13	94.19	96.08
SAFDNet [65]	2025	99.55	90.73	95.14	94.47	95.82
CSDNet	2025	99.56	90.88	95.22	95.12	95.33

TABLE IV
COMPARISON OF DIFFERENT METHODS ON THE MSRSCD DATASET

Method	Year	OA	IoU	F1	Rec	Prec
STANet [45]	2020	91.44	54.52	70.57	69.46	71.71
SGSLN [57]	2023	92.52	56.28	73.36	69.73	77.39
ChangeFormer [22]	2022	91.86	56.96	72.58	72.94	72.22
FCCDN [70]	2021	92.36	57.94	73.37	71.30	75.56
AANet [71]	2024	92.17	59.23	74.40	77.03	71.94
EATDer [72]	2024	91.47	59.29	74.44	84.17	66.73
MSNet [68]	2025	93.03	59.80	75.74	74.73	76.79
DMINet [61]	2023	93.05	61.24	75.96	74.37	77.63
BASNet [53]	2024	92.87	61.31	76.02	76.51	75.53
CDNeXt [51]	2024	93.24	61.59	76.23	73.42	79.27
DFFNet [73]	2025	93.23	61.82	77.44	78.69	76.22
CSDNet	2025	93.07	62.01	76.55	76.38	76.73

TABLE V
COMPARISON OF DIFFERENT METHODS ON THE LEVIR-CD DATASET

Method	Year	OA	IoU	F1	Rec	Prec
BIT [7]	2022	98.95	80.86	89.48	87.53	90.65
STANet [45]	2020	99.02	81.85	90.02	87.13	93.10
ChangeFormer [22]	2022	99.04	82.66	90.50	90.18	90.83
P2V [47]	2023	99.04	83.00	90.71	91.78	89.67
CDMamba [74]	2024	99.06	83.07	90.75	90.08	91.43
CDxLSTM [75]	2024	99.07	83.30	90.89	90.27	91.52
MSNet [68]	2025	99.14	83.60	91.41	89.99	92.87
ISDANet [56]	2025	99.10	83.63	91.06	90.08	92.19
STRobustNet [76]	2025	99.10	83.66	91.11	90.67	91.54
B2CNet [77]	2024	99.10	83.72	91.14	91.01	91.27
DgFA [78]	2025	99.11	83.93	91.26	90.84	91.69
FSG-Net [79]	2025	99.10	83.94	91.27	90.04	92.53
ChangeCLIP [34]	2024	99.14	83.99	91.30	89.04	93.68
HATNet [54]	2024	99.13	84.12	91.38	90.15	92.64
STransUNet [80]	2022	99.13	84.19	91.41	90.55	92.30
MDENet [81]	2025	99.16	84.21	91.43	90.27	92.61
CSDNet	2025	99.16	84.47	91.58	90.00	93.23

posed by each dataset, which our CSDM and CCRM modules are designed to address.

We provide a deeper analysis linking our model's effectiveness to the specific characteristics of each dataset as follows.

1) *On LEVIR-CD:* This dataset features a long time span (5–14 years), resulting in significant style discrepancies from seasonal, lighting, and sensor variations. This is a primary challenge for many models, leading to high false positives. CSDNet's CSDM module directly confronts this by decoupling and normalizing these style differences in the encoder. This allows the model to focus on the "content" (i.e., the presence or absence of buildings) rather than spurious "style" changes (e.g., roof color, shadows, and vegetation state), leading to a robust IoU of 84.47.
2) *On S2Looking:* This is one of the most challenging datasets in change detection due to its strong seasonal interference from vast rural farmland, oblique (side-on) viewing angles, and sparse building targets. Standard models often mistake seasonal farmland changes for true changes. CSDNet's superior performance (IoU 50.72) demonstrates the CSDM's ability to aggressively suppress this dominant, irrelevant seasonal "style" noise, while also normalizing the unusual geometric and illumination patterns from oblique imagery. This effectively isolates the sparse, true building changes.
3) *On WHUCD:* This dataset's challenge is its extremely high resolution (0.075–0.3 m). While this provides rich detail, it also amplifies interference from fine-grained roof textures and complex shadows, making precise boundary delineation difficult. Here, our CCRM module plays a critical role. In the decoding stage, CCRM's joint channel–spatial gating mechanism refines the fused features, sharpening the model's focus on the true structural edges of buildings while filtering out shadow-related noise. This leads to the highest IoU (90.88) and a well-balanced precision/recall.
4) *On SYSU-CD and MSRSCD:* These "general-purpose" datasets test a model's generalization across diverse and complex backgrounds (e.g., urban, suburban, vegetation, and water). CSDNet's top performance (IoU 71.16 on SYSU and IoU 62.01 on MSRSCD) highlights the synergistic effect of CSDM and CCRM. CSDM provides a robust, style-agnostic feature representation from the varied inputs, and CCRM ensures the final decoded feature map is contextually refined. This dual-stage processing makes CSDNet highly robust to the heterogeneous scenarios present in these datasets.

Figs. 3–7 present qualitative comparisons for the SYSU-CD, S2Looking, WHUCD, and MSRSCD datasets. To facilitate intuitive interpretation, different colors represent true positives (TPs, white), false positives (FPs, red), true negatives (TNs, black), and false negatives (FNs, green). In challenging samples with complex land-cover changes, CSDNet exhibits fewer false positives and false negatives than other methods, and its boundary localization is particularly accurate on small-scale changes, as observed in SYSU-CD and WHUCD. Models such as HATNet and STANet show more frequent false positives, while CDNeXt occasionally misses subtle changes. These visual results corroborate our quantitative analysis: the reduction in false positives (red) in S2Looking and LEVIR-CD samples confirms CSDM's style suppression capability. Meanwhile, the clean boundaries (white) in WHUCD samples validate CCRM's refinement effect. These observations align with the quantitative results, confirming that CSDNet effectively suppresses irrelevant style noise and emphasizes change-relevant structural information.

F. Computational Cost Analysis With Various Backbones

As shown in Table VII, we conduct a controlled backbone-replacement study where only the feature encoder is swapped,

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TABLE VI
RESEARCH FOCUSES OF COMPARISON METHODS

Research Focuses	Models
Transformer	BIT [7], EATDer [72], ChangeFormer [22], ChangeCLIP [34], STransUNet [80], DgFA [78]
Attention	CDNeXt [51], STANet [45], HATNet [54], BASNet [53], ISDANet [56], B2CNet [77], DMATNet [26], STRobustNet [76]
Difference Feature Modeling	CGNet [49], HFIFNet [63], EfficientCD [58], P2V [47], FHD [59], Changer [64], DMINet [61], MDENet [81], FSG-Net [79]
AI Foundational Models	ChangeCLIP [34], SAM-CD [32], AdaptVFMs-RSCD [55]
Mamba	ChangeMamba [29], VMMCD [60], CDMamba [74]
Multi Scale Feature Refine	DFFNet [73], AANet [71], PCAANet [62], SNUNet [6], HCGMNet [48], DARNet [50], SAFDNet [65], MSNet [68], FCCDN [70], L-UNet [46], SGSLN [57], CDxLSTM [75], STDFCD [52]

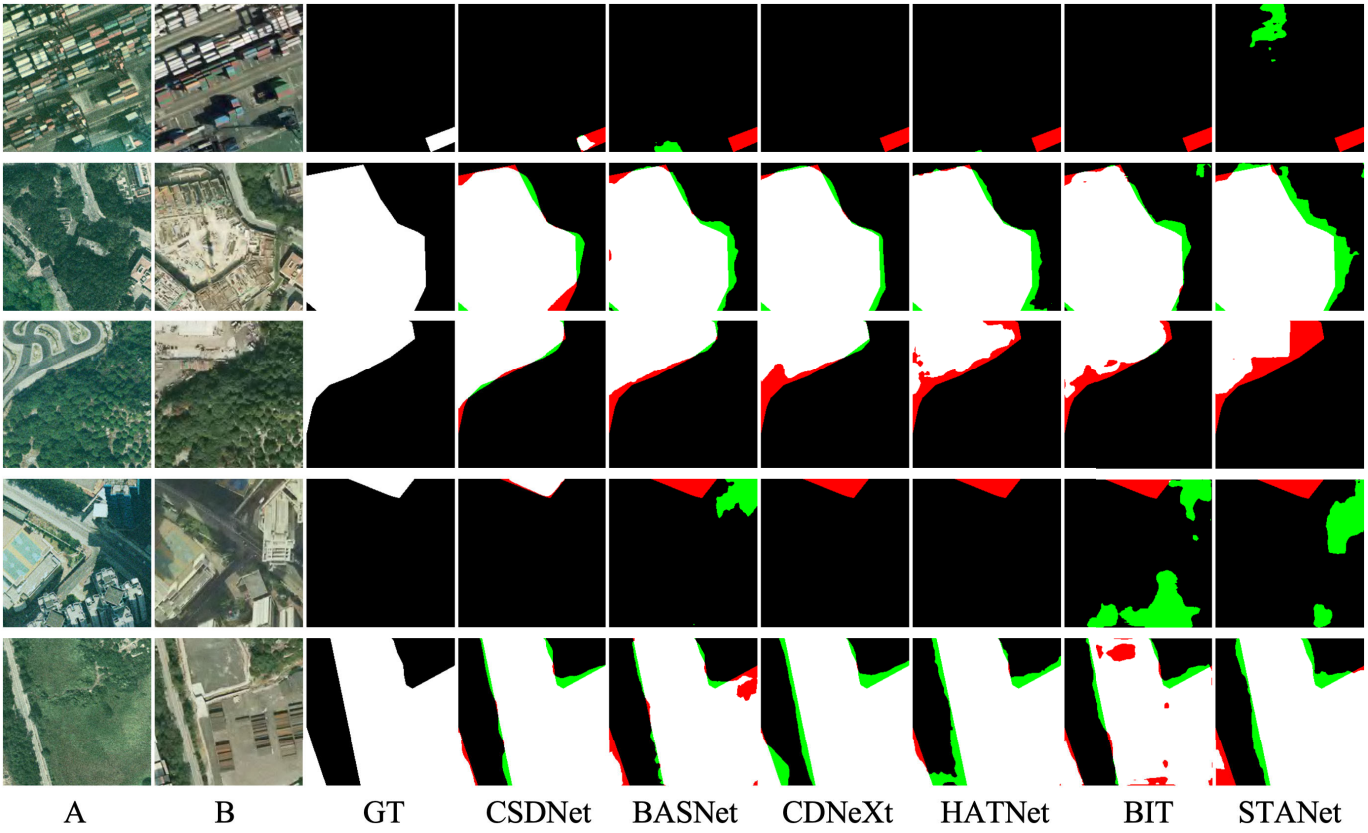


Fig. 3. Visualization comparison of different methods on the SYSU-CD test set.

while all other components (CSDM/CCRM, FPN, decoder, and training protocol) are kept unchanged. This analysis is intended to verify the effectiveness and backbone-agnostic nature of the proposed modules rather than to optimize throughput or latency; accordingly, no pruning, quantization, or kernel-level acceleration is applied.

On SYSU-CD, ResNet18 [2] (20.23-M parameters, 64.21-G FLOPs) attains the best IoU (72.30) and $F1$ (83.92), exceeding or matching larger backbones such as SwinTv2-Base [82] (75.43 M, 115.33 G; IoU 70.13) and ConvNeXt-Base [83] (97.01 M, 135.41 G; IoU 63.38). Lightweight CNN/transformer variants (EfficientNet-B0 [84], HRNet18 [13], SwinTv2-Tiny [82]) consistently achieve IoU around 70 ± 1 with markedly lower complexity (56–74-G FLOPs). On WHUCD, MambaOut-Tiny/Base delivers strong IoU (92.16/92.39) and $F1$ (95.92/96.05), comparable to or better than larger CNN/transformer counterparts, while HRNet18/32 remain competitive. The reference configuration HRNet64 (CSDNet) exhibits stable performance (SYSU-CD IoU 71.16; WHUCD IoU 90.88) but also the highest FLOPs, reflecting a design preference for robust multiscale representation rather than efficiency.

These observations indicate that the gains of CSDNet primarily stem from the proposed content–style disentanglement with gated selection, rather than from increased model capacity. Notably, comparable accuracy can be obtained

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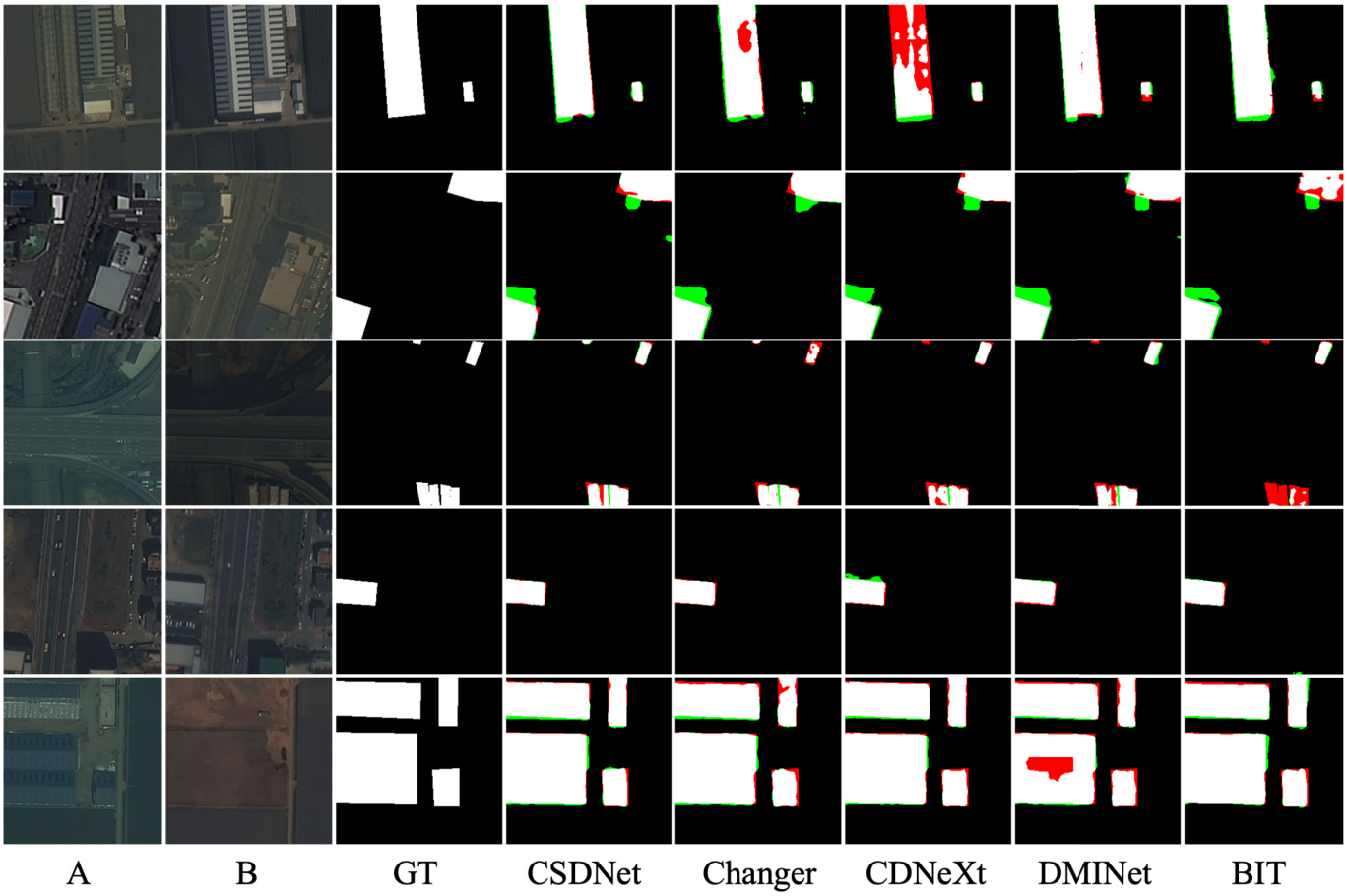


Fig. 4. Visualization comparison of different methods on the S2Looking test set.

TABLE VII
PERFORMANCE COMPARISON ON SYSU-CD AND WHUCD

Backbone	Params (M)	FLOPs (G)	SYSU-CD			WHUCD		
			IoU	F1	Rec	IoU	F1	Rec
SwinTv2-Tiny	30.36	72.42	70.57	82.75	78.22	91.49	95.56	94.28
SwinTv2-Small	46.42	88.87	69.60	82.08	76.43	91.78	95.71	94.45
ConvNeXt-Tiny	37.06	78.22	70.03	82.37	78.53	91.36	95.49	94.71
ResNet18	20.23	64.21	72.30	83.92	82.23	87.96	93.60	94.45
EfficientNet-B0	12.45	56.27	70.00	82.36	76.75	89.14	94.26	94.14
MambaOut-Tiny	32.05	78.26	68.97	81.63	78.69	92.16	95.92	95.03
HRNet18	20.45	73.74	70.65	82.80	81.68	90.10	94.79	94.23
MobileNetV4-Small	10.45	55.43	69.19	81.79	76.89	84.49	91.59	93.68
ResNet50	34.29	77.81	70.48	82.68	80.99	89.14	94.26	94.14
EfficientNet-B5	36.12	66.52	69.90	82.28	78.50	91.38	95.50	94.92
HRNet32	40.38	98.10	70.34	82.59	79.68	90.45	94.98	94.70
ConvNeXt-Base	97.01	135.41	63.38	77.59	74.56	92.31	96.00	94.62
MambaOut-Base	88.68	137.67	69.21	81.80	75.33	92.39	96.05	94.73
SwinTv2-Base	75.43	115.33	70.13	82.44	76.95	91.66	95.65	94.30
HRNet64 (CSDNet)	127.21	202.74	71.16	83.15	79.60	90.88	95.22	95.12

with lightweight backbones without any architecture-specific tuning, suggesting that CSDM/CCRM act as plug-and-play, architecture-agnostic modules for suppressing nuisance style while preserving task-relevant cues. Consequently, efficiency improvements are orthogonal to our contribution and can be pursued by combining CSDNet with standard acceleration techniques when required by deployment constraints.

G. Ablation Study in CSDNet

To validate the effectiveness of the proposed CSDM and contextual content refiner module (CCRM), we conducted a series of ablation experiments, evaluating the performance of the model under different configurations on five public datasets: SYSU-CD, LEVIR-CD, WHUCD, MSRSCD, and S2Looking, with IoU as the primary

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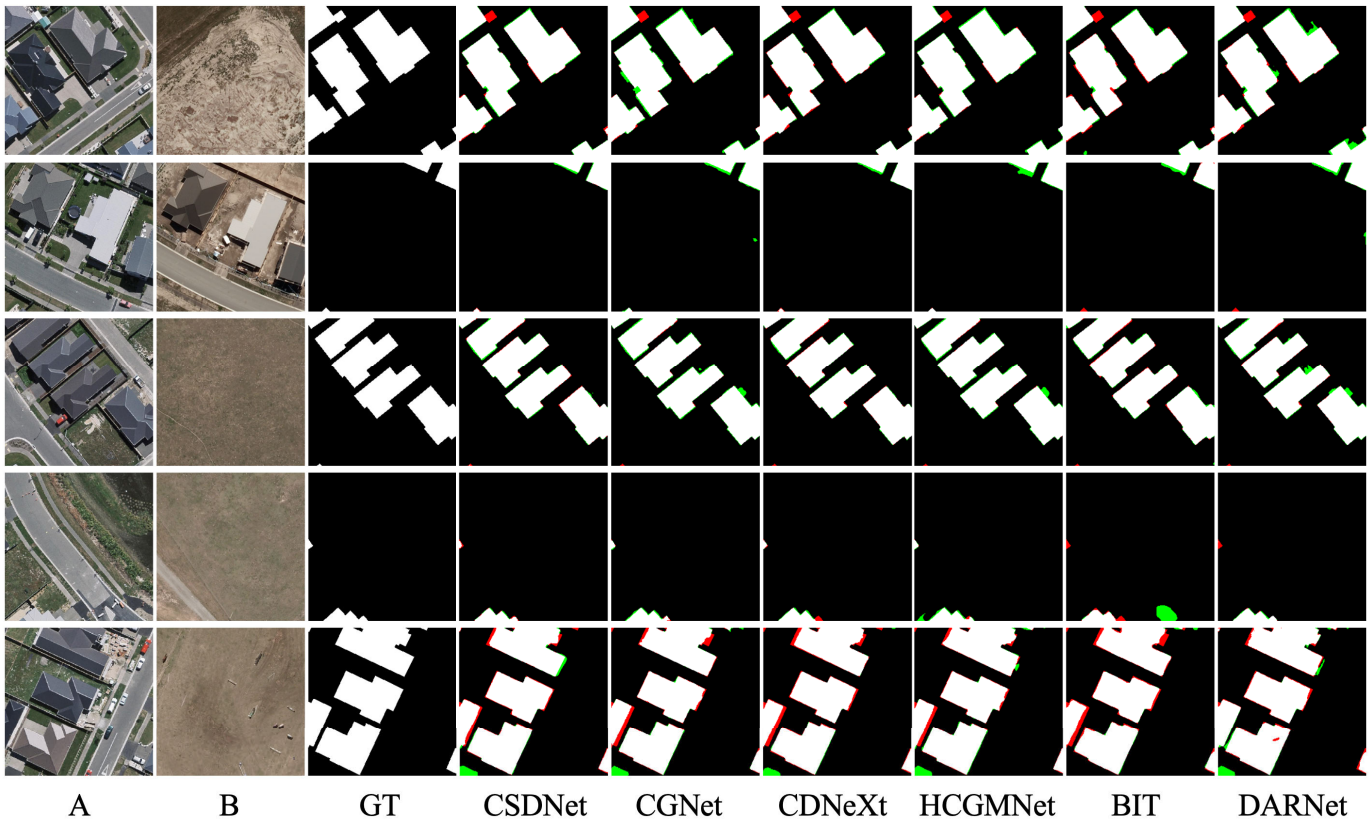


Fig. 5. Visualization comparison of different methods on the WHUCD test set.

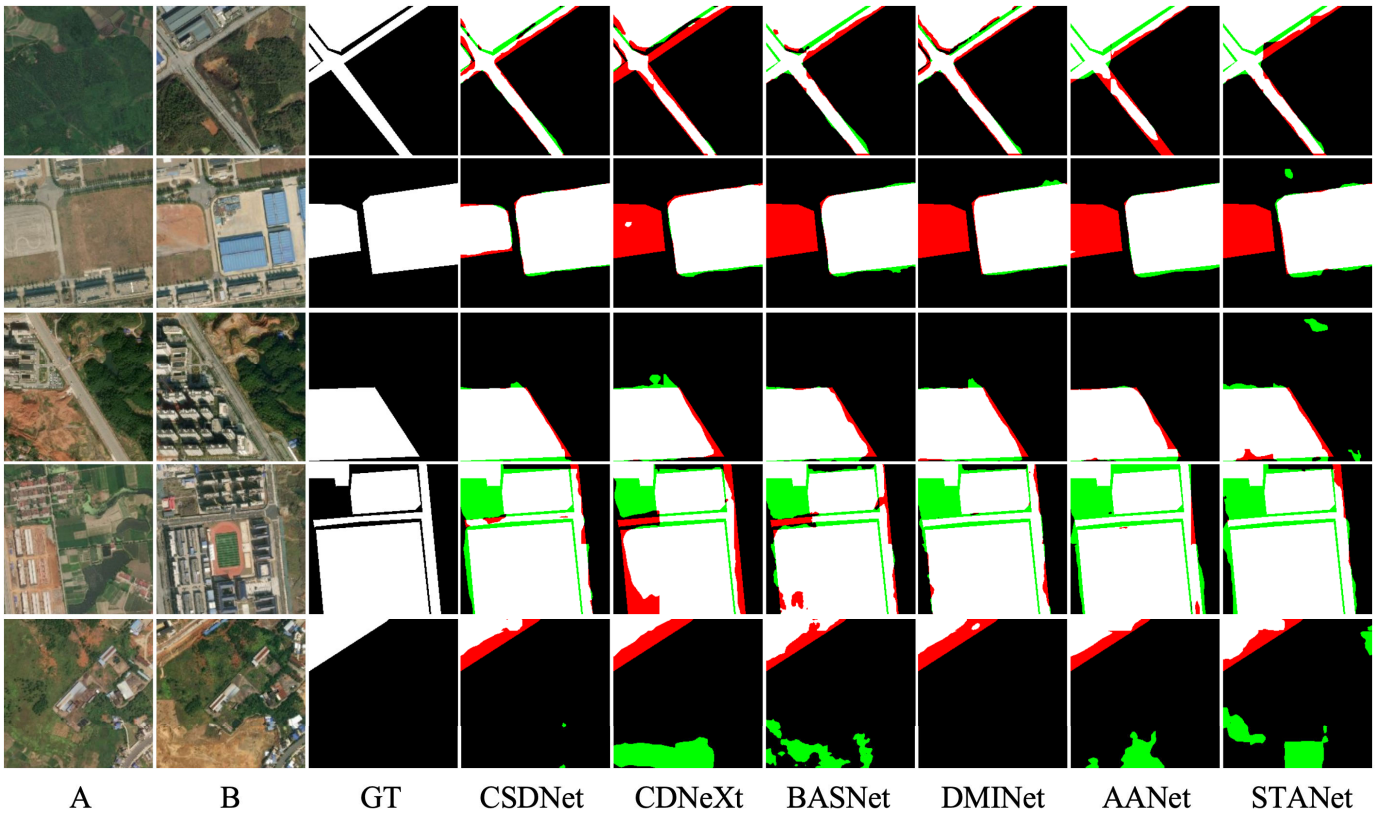


Fig. 6. Visualization comparison of different methods on the MSRSCD test set.

evaluation metric. As shown in Table VIII, when the model did not include the CSDM and CCRM modules, serving as a baseline, its IoU performance across the datasets was at a foundational level. Encouragingly,

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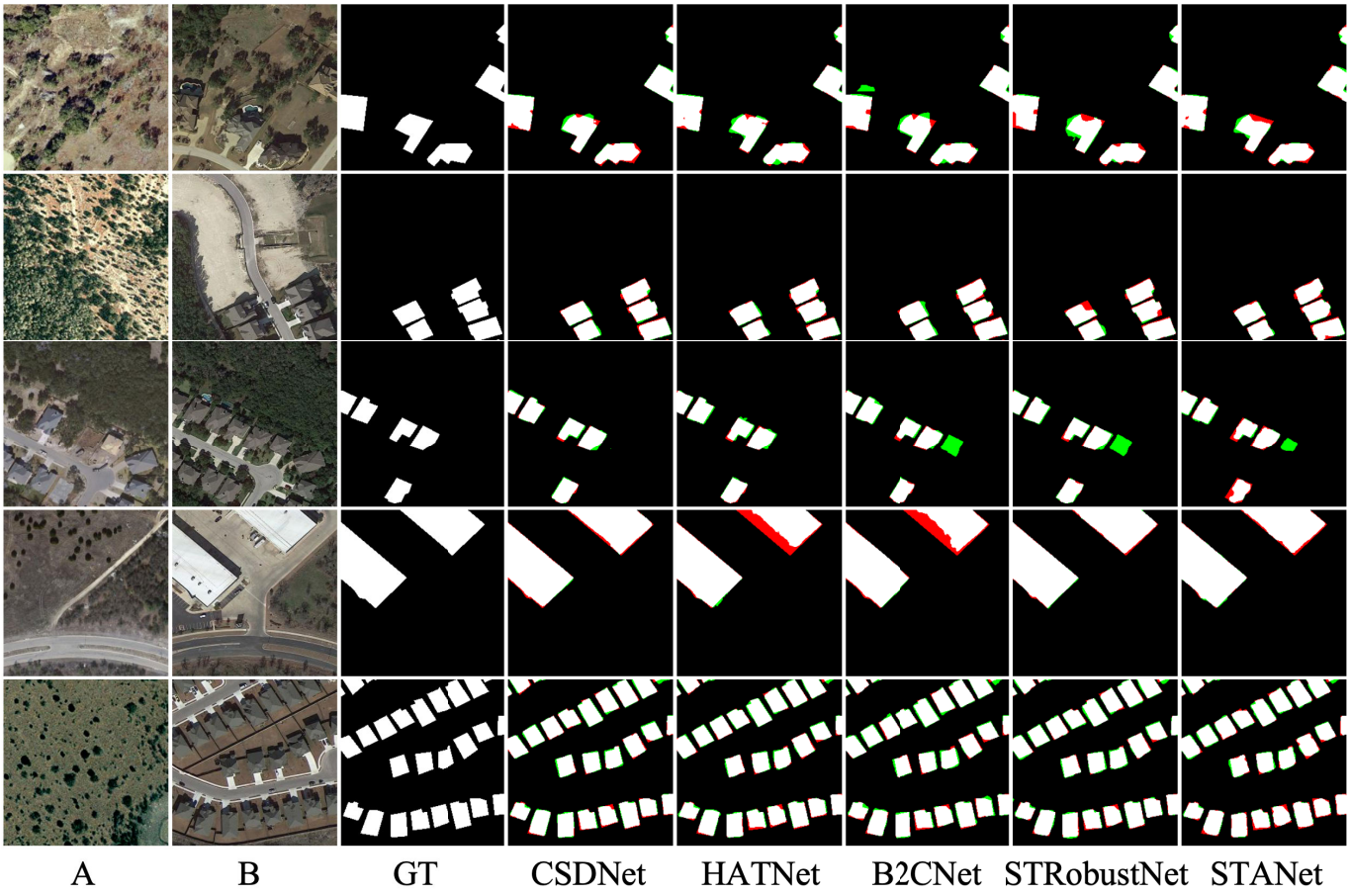


Fig. 7. Visualization comparison of different methods on the LEVIR-CD test set.

TABLE VIII
ABLATION STUDY RESULTS ON DIFFERENT DATASETS

CSDM	CCRM	SYSU-CD			LEVIR-CD			WHUCD			MSRSCD			S2Looking		
		IoU	F1	Rec	IoU	F1	Rec	IoU	F1	Rec	IoU	F1	Rec	IoU	F1	Rec
×	×	70.30	82.56	79.05	83.62	91.08	89.62	89.78	94.61	94.53	59.95	74.96	74.81	49.88	66.56	63.51
✓	×	70.59	82.76	79.21	83.94	91.27	89.81	90.22	94.85	94.76	61.46	76.12	76.03	50.31	66.94	63.99
×	✓	70.64	82.83	79.24	83.85	91.21	89.72	90.38	94.94	94.88	60.88	75.69	75.52	50.22	66.86	63.85
✓	$CCRM \rightarrow CBAM$	70.41	82.64	81.05	83.97	91.28	89.53	87.42	93.29	94.49	61.29	76.00	71.76	48.62	65.43	60.87
✓	$CCRM \rightarrow SE$	69.45	81.97	77.70	83.61	91.07	89.31	89.39	94.40	94.70	60.81	75.63	71.76	48.94	65.72	58.58
✓	$CCRM \rightarrow NL$	70.84	82.93	78.76	83.69	91.12	88.83	90.57	95.05	94.49	61.42	76.10	71.78	49.81	66.38	63.05
✓	✓	71.16	83.15	79.60	84.47	91.58	90.00	90.88	95.22	95.12	62.01	76.55	76.38	50.72	67.31	64.34

TABLE IX
DOMAIN DISCREPANCY METRICS FROM UMAP EMBEDDINGS BEFORE AND AFTER TRAINING ON FIVE DATASETS. NEGATIVE DELTA MEANS THE METRIC DECREASED AFTER TRAINING (TYPICALLY INDICATING REDUCED DOMAIN GAP)

Metric	Stage	WHUCD	MSRSCD	S2Looking	LEVIR-CD	SYSU-CD
Centroid dist	Before training	9.5444	1.0438	8.1495	7.2442	13.0549
	After training	7.8133	1.2386	7.5414	2.8456	11.2671
	Δ	**-1.7311**	**+0.1948**	**-0.6081**	**-4.3986**	**-1.7878**
MMD	Before training	0.1655	0.0526	0.1053	0.2022	0.3706
	After training	0.1213	0.0560	0.0953	0.1229	0.1448
	Δ	**-0.0442**	**+0.0034**	**-0.0100**	**-0.0793**	**-0.2258**
CORAL	Before training	821.4465	384.3917	1941.6185	3277.2930	820.3972
	After training	1502.2324	326.6985	1338.1335	1628.3154	451.8247
	Δ	**+680.7859**	**-57.6932**	**-603.4850**	**-1648.9776**	**-368.5725**

upon the independent introduction of the CSDM module, the model's performance significantly improved across all datasets. This strongly demonstrates that CSDM, through its content–style disentanglement mechanism, effectively

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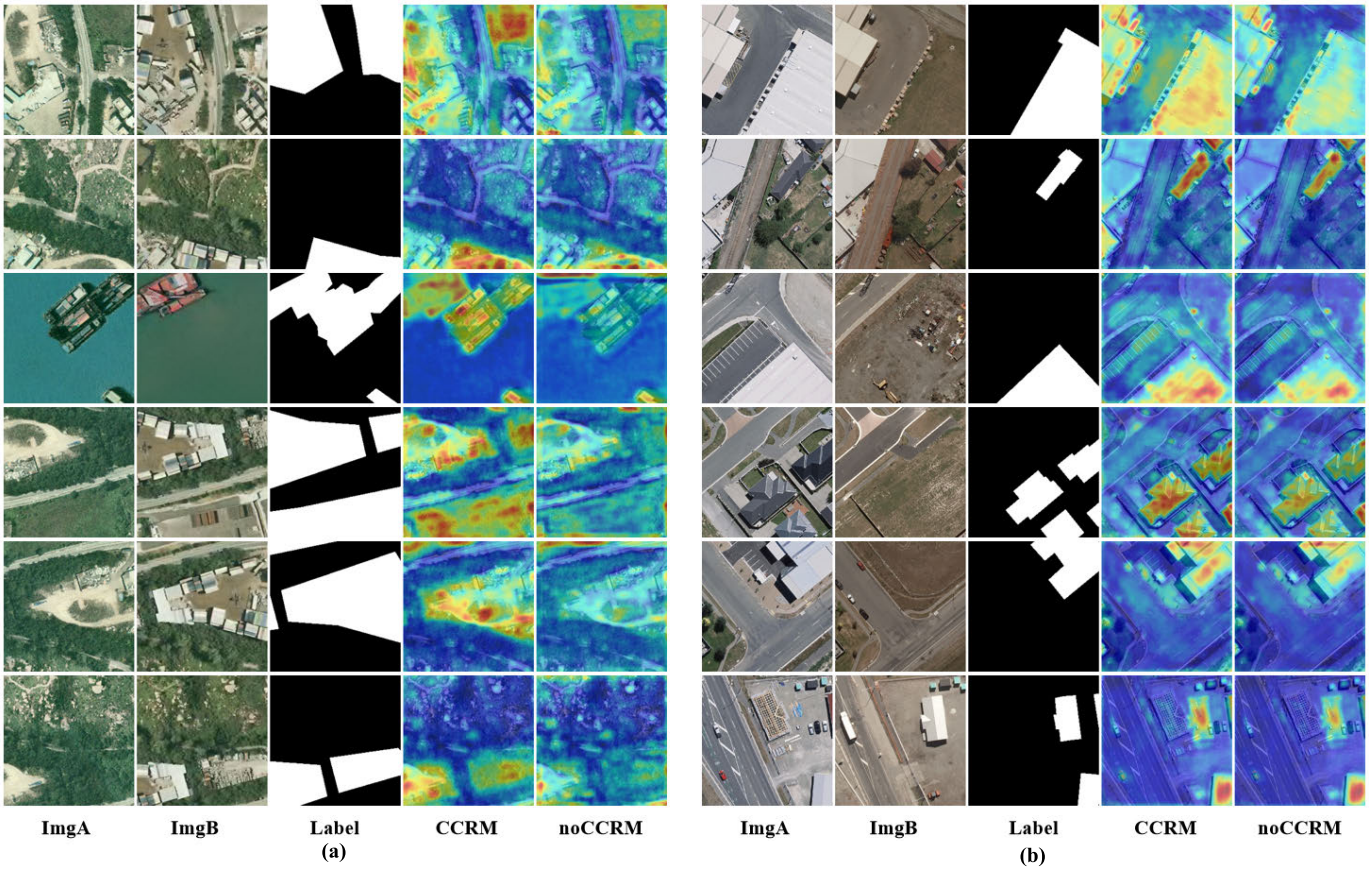


Fig. 8. Comparison of heatmap visualizations of the CCRM module on (a) SYSU-CD and (b) WHUCD datasets. From left to right: 1) T1 image, 2) T2 image, 3) ground truth, 4) CCRM, and 5) no CCRM. The heatmaps highlight the regions where the CCRM module focuses its attention during the refinement process, demonstrating its effectiveness in enhancing change detection performance.

suppresses background style noise and enhances feature alignment between bitemporal images, thereby improving the model's ability to perceive real changes, showcasing particular robustness when handling images with complex or heterogeneous style differences. Similarly, with the independent introduction of the CCRM module, the model's performance also showed positive improvements. This validates that CCRM, in the decoding stage, can further refine features through contextual content refinement and a joint channel–spatial gating mechanism, enhancing the perception of change regions and contributing to more fine-grained delineation of changed areas. Most importantly, when both the CSDM and CCRM modules were integrated simultaneously, the model's performance reached its optimum, achieving the highest IoU across all datasets. This powerfully demonstrates that CSDM and CCRM do not simply provide additive effects but rather enhance the model's feature representation and change detection capabilities through effective synergistic work in both the encoding and decoding stages. CSDM processes style noise and purifies features in the early stage, while CCRM performs more fine-grained context-aware refinement on these purified features in the later stage.

V. Discussion

A. Content–Style Disentanglement and Change Sensitivity

The CSDM demonstrates empirical effectiveness in mitigating style interference in bitemporal images and enhancing change sensitivity. As shown in Table IX, we have calculated three domain discrepancy metrics, centroid distance, maximum mean discrepancy (MMD), and CORAL, on the UMAP embedding space of the features before and after training. These metrics quantify the alignment of feature distributions between the bitemporal images, with lower values indicating better alignment and reduced style noise.

The introduction of CSDM leads to a notable decrease in both centroid distance and MMD within the UMAP embedding space across most datasets. Specifically, on SYSU CD, the centroid distance drops from 13.0549 to 11.2671 and MMD from 0.3706 to 0.1448. These results indicate that the discrepancy between the dual temporal feature distributions is compressed at both the geometric center level and within the reproducing kernel space, effectively suppressing shifts in style and domain. Furthermore, the CORAL metric exhibits a downward trend on datasets such as MSRSCD, S2Looking, and LEVIR CD, reflecting a more consistent reconstruction of the covariance structure following disentanglement. The observed increase in CORAL on WHUCD may suggest that CSDM removes explicit style noise while structurally reshaping the original covariance to preserve or adapt essential fine-grained statistical information. Overall, these quantitative changes support the design intent of CSDM: through InstanceNorm and channel-wise gating, it selectively removes nuisance style factors (e.g., illumination, seasonality, and sensor noise) while retaining beneficial style cues such as edges and textures, thereby improving the model's discriminability for real changes. The module not only achieves better alignment in feature representation but also yields more robust change detection performance.

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B. Contextual Content Refinement and Boundary Clarity

Beyond the encoder-side suppression of cross-temporal style noise by CSDM, the decoder-side CCRM is introduced to conduct fine-grained cleansing and reorganization of features from the postfusion distribution. Concretely, we first perform a lightweight content–style redecomposition on the fused feature P, i.e., $C = \mathrm{IN}(P)$ and $S = P - C$. We then apply dual gating along the channel and spatial dimensions only to the style residual S to obtain S', and finally conduct content-preserving selective reinjection via $\hat{P} = C + S'$ to form a cleaner reconstructed representation. This style-targeted, content-anchored modulation suppresses pseudo textures, shadows, and illumination residues introduced by FPN fusion and upsampling—without weakening structural semantics—thereby improving separability at decision-critical regions (object boundaries and small targets).

To substantiate this mechanism, we add heatmap visualizations before and after CCRM, as shown in Fig. 8. We select representative samples from SYSU-CD and WHUCD, and, by columns, present the bitemporal pair (ImgA/ImgB), the ground-truth (Label), the response with CCRM, and the response before enabling CCRM. The comparison shows that: 1) under complex backgrounds (e.g., tidal flats, bare soil, road networks, and roof shadows), CCRM markedly suppresses large low-frequency "hot" areas, preventing pseudo differences from being amplified in the decoder; 2) near object boundaries, the spatial gating of CCRM produces responses that are tighter and thinner along the edges, with energy better concentrated and more consistent with the ground-truth contours; and 3) for small-scale structures (e.g., vessels and roof details), CCRM preserves the necessary textural cues without over-spilling into the background. These observations align with our design: CCRM constrains attention within the style-residual subspace and, through selective reinjection, retains only style cues that benefit change discrimination while avoiding over-modulation of the content backbone.

From a system perspective, CSDM and CCRM constitute a two-stage adaptation of global precleaning followed by local postrefinement: the former aligns cross-temporal/cross-domain statistics at the encoder side, while the latter performs conditional refinement of the posterior style residuals induced by fusion and resampling right before prediction. This synergy reduces both false positives and false negatives while yielding sharper and more complete boundaries. It is consistent with our quantitative results and ablations in Table VIII for the comparisons among CCRM and variants "*CCRM* → CBAM [85]/SE [86]/NL [87]"), where we observe stable gains.

VI. Conclusion

This article proposes a bitemporal change detection network, CSDNet, based on content–style disentanglement and channel–spatial gating. In the encoding stage, the network uses the CSDM to achieve InstanceNorm and style residual extraction on multiscale features, and utilizes a channel gate to adaptively preserve style cues beneficial for change detection. In the decoding stage, the contextual content refiner module (CCRM) performs dual-channel and spatial gating refinement on the style residuals before recombining them with content features, significantly improving the clarity of change boundaries and detection accuracy. Experimental results show that CSDNet outperforms various state-of-the-art methods on the LEVIR-CD, SYSU-CD, WHUCD, S2Looking, and MSRSCD datasets in terms of $F1$, IoU, and precision metrics, validating the high robustness and accuracy of the proposed modules in complex land-cover backgrounds.

Acknowledgment

The numerical calculations in this article have been done on the supercomputing system in the Supercomputing Center of Wuhan University.

References

[1] D. Sun, J. Yao, W. Xue, C. Zhou, P. Ghamisi, and X. Cao, "Mask approximation net: A novel diffusion model approach for remote sensing change captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5652311.

[2] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[3] A. Vaswani et al., "Attention is all you need," in *Proc. Neural Inf. Process. Syst.*, vol. 30, 2017.

[4] G. R. Koch. (2015). *Siamese Neural Networks for One-Shot Image Recognition*. [Online]. Available: https://api.semanticscholar.org/CorpusID:13874643

[5] J. Chen et al., "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, Nov. 2021. [Online]. Available: https://api.semanticscholar.org/CorpusID:212633986

[6] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022. [Online]. Available: https://api.semanticscholar.org/CorpusID:233967703

[7] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," 2021, *arXiv:2103.00208*.

[8] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224713. [Online]. Available: https://ieeexplore.ieee.org/document/9736956/

[9] L. McInnes, J. Healy, N. Saul, and L. Großberger, "UMAP: Uniform manifold approximation and projection," *J. Open Source Softw.*, vol. 3, no. 29, p. 861, Sep. 2018. [Online]. Available: https://api.semanticscholar.org/CorpusID:53244226

[10] M. Cheng, W. He, Z. Li, G. Yang, and H. Zhang, "Harmony in diversity: Content cleansing change detection framework for very-high-resolution remote-sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 218, pp. 1–19, Dec. 2024.

[11] X. Jiang, G. Li, Y. Liu, X.-P. Zhang, and Y. He, "Change detection in heterogeneous optical and SAR remote sensing images via deep homogeneous feature fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1551–1566, 2020.

[12] J. Zhang et al., "Frequency decoupled domain-irrelevant feature learning for pan-sharpening," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 2, pp. 1237–1250, Feb. 2025. [Online]. Available: https://api.semanticscholar.org/CorpusID:273393904

[13] J. Wang et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2021. [Online]. Available: https://api.semanticscholar.org/CorpusID:201124533

[14] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944. [Online]. Available: https://api.semanticscholar.org/CorpusID:10716717

[15] G. Ceng, G. Wang, and J. Han, "Deep learning for change detection in remote sensing: A review and new outlooks," *Nat. Remote Sens. Bull.*, vol. 29, no. 6, pp. 1587–1597, Jun. 2025. [Online]. Available: https://api.semanticscholar.org/CorpusID:277989318

[16] S. Mishra, P. Shrivastava, and P. Dhurvey, "Change detection techniques in remote sensing: A review," *Int. J. Wireless Mobile Commun. Ind. Syst.*, vol. 4, no. 1, pp. 1–8, 2017. [Online]. Available: https://api.semanticscholar.org/CorpusID:133632415

[17] Z. Lv et al., "Land cover change detection with heterogeneous remote sensing images: Review, progress, and perspective," *Proc. IEEE*, vol. 110, no. 12, pp. 1976–1991, Dec. 2022. [Online]. Available: https://api.semanticscholar.org/CorpusID:253695791

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[18] C. Wu, L. Zhang, B. Du, H. Chen, J. Wang, and H. Zhong, "UNet-like remote sensing change detection: A review of current models and research directions," *IEEE Geosci. Remote Sens. Mag. (replaces Newsletter)*, vol. 12, no. 4, pp. 305–334, Dec. 2024. [Online]. Available: https://api.semanticscholar.org/CorpusID:271624991

[19] Z. Zhou, M. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-Net architecture for medical image segmentation," in *Proc. Int. Workshop Deep Learn. Med. Image Anal.*, vol. 11045, 2018, pp. 3–11.

[20] S. Zhu, Y. Song, Y. Zhang, and Y. Zhang, "ECFNet: A Siamese network with fewer FPs and fewer FNs for change detection of remote-sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[21] X. Qu, F. Gao, J. Dong, Q. Du, and H.-C. Li, "Change detection in synthetic aperture radar images using a dual-domain network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[22] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[23] Y. Tang, Z. Cao, N. Guo, and M. Jiang, "A Siamese Swin-UNet for image change detection," *Sci. Rep.*, vol. 14, no. 1, p. 4577, Feb. 2024.

[24] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[25] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[26] X. Song, Z. Hua, and J. Li, "Remote sensing image change detection transformer network based on dual-feature mixed attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5920416. [Online]. Available: https://api.semanticscholar.org/CorpusID:252605199

[27] M. Liu, Q. Shi, Z. Chai, and J. Li, "PA-former: Learning prior-aware transformer for remote sensing building change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[28] X. Zhang, S. Cheng, L. Wang, and H. Li, "Asymmetric cross-attention hierarchical network based on CNN and transformer for bitemporal remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2000415.

[29] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409720.

[30] A. Kirillov et al., "Segment anything," in *Proc. Int. Conf. Comput. Vision (ICCV)*, Mar. 2023, pp. 3992–4003. [Online]. Available: https://api.semanticscholar.org/CorpusID:257952310

[31] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, Jul. 2021, pp. 8748–8763. [Online]. Available: https://api.semanticscholar.org/CorpusID:231591445

[32] L. Ding, K. Zhu, D. Peng, H. Tang, K. Yang, and L. Bruzzone, "Adapting segment anything model for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611711.

[33] Z. Zheng, Y. Zhong, L. Zhang, and S. Ermon, "Segment any change," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 37, 2024, pp. 81204–81224.

[34] S. Dong, L. Wang, B. Du, and X. Meng, "ChangeCLIP: Remote sensing change detection with multimodal vision-language representation learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, pp. 53–69, Feb. 2024.

[35] K. Chen et al., "Time travelling pixels: Bitemporal features integration with foundation model for remote sensing image change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2024, pp. 8581–8584.

[36] K. Li, X. Cao, and D. Meng, "A new learning paradigm for foundation model-based remote-sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610112.

[37] M. Wang et al., "RSBuilding: Toward general remote sensing image building extraction and change detection with foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4707417.

[38] F. Zhou, X. Zhang, H. Shuai, Q. Liu, and R. Hang, "Adaptive frequency collaboration for remote sensing change detection," *Neural Netw.*, vol. 195, Mar. 2026, Art. no. 108234. [Online]. Available: https://api.semanticscholar.org/CorpusID:282215372

[39] Y. Tu, L. Li, L. Su, J. Du, K. Lu, and Q. Huang, "Viewpoint-adaptive representation disentanglement network for change captioning," *IEEE Trans. Image Process.*, vol. 32, pp. 2620–2635, 2023. [Online]. Available: https://api.semanticscholar.org/CorpusID:258311232

[40] B. Wang, X. Jiang, P. Qin, and J. Zeng, "DSFDcd: Joint distribution sampling and feature decoupling deep network for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5631312. [Online]. Available: https://api.semanticscholar.org/CorpusID:279880408

[41] X. Li, Z. Du, Y. Huang, and Z. Tan, "A deep translation (GAN) based change detection network for optical and SAR remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 179, pp. 14–34, Sep. 2021.

[42] H. Chen, Y. Zao, L. Liu, S. Chen, and Z. Shi, "Semantic decoupled representation learning for remote sensing image change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 1051–1054.

[43] B. Fang, G. Chen, G. Ouyang, J. Chen, R. Kou, and L. Wang, "Content-invariant dual learning for change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5603317. [Online]. Available: https://api.semanticscholar.org/CorpusID:234107567

[44] A. Dai, J. Yang, T. Zhang, B. Gao, K. Tang, and X. Chen, "DADR-HCD: A deep domain adaptation and disentangled representation network for unsupervised heterogeneous change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5625315. [Online]. Available: https://api.semanticscholar.org/CorpusID:269963824

[45] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020. [Online]. Available: https://www.mdpi.com/2072-4292/12/10/1662

[46] M. Papadomanolaki, M. Vakalopoulou, and K. Karantzalos, "A deep multitask learning framework coupling semantic segmentation and fully convolutional LSTM networks for urban change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7651–7668, Sep. 2021. [Online]. Available: https://api.semanticscholar.org/CorpusID:233910177

[47] M. Lin, G. Yang, and H. Zhang, "Transition is a process: Pair-to-video change detection networks for very high resolution remote sensing images," *IEEE Trans. Image Process.*, vol. 32, pp. 57–71, 2023. [Online]. Available: https://ieeexplore.ieee.org/document/9975266/

[48] C. Han, C. Wu, and B. Du, "HCGMNet: A hierarchical change guiding map network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Pasadena, CA, USA, Jul. 2023, pp. 5511–5514. [Online]. Available: https://api.semanticscholar.org/CorpusID:257050289

[49] C. Han, C. Wu, H. Guo, M. Hu, J. Li, and H. Chen, "Change guiding network: Incorporating change prior to guide change detection in remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 8395–8407, 2023.

[50] Z. Li, C. Yan, Y. Sun, and Q. Xin, "A densely attentive refinement network for change detection based on very-high-resolution bitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4409818. [Online]. Available: https://ieeexplore.ieee.org/document/9734050/

[51] J. Wei et al., "Robust change detection for remote sensing images based on temporospatial interactive attention module," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 128, Apr. 2024, Art. no. 103767.

[52] Y. Zhou, S. Zhou, D. Cheng, J. Li, and Z. Hua, "STDF: Joint spatiotemporal differences based on xLSTM dendritic fusion network for remote sensing change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 1–16, 2025. [Online]. Available: https://api.semanticscholar.org/CorpusID:277291037

[53] Z. Wang, G. Gu, M. Xia, L. Weng, and K. Hu, "Bitemporal attention sharing network for remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 10368–10379, 2024.

[54] C. Xu et al., "Hybrid attention-aware transformer network collaborative multiscale feature alignment for building change detection," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–14, 2024. [Online]. Available: https://api.semanticscholar.org/CorpusID:268375803

[55] W. Jiang, Y. Sun, L. Lei, G. Kuang, and K. Ji, "AdaptVFMs-RSCD: Advancing remote sensing change detection from binary to semantic with SAM and CLIP," *ISPRS J. Photogramm. Remote Sens.*, vol. 230, pp. 304–317, Dec. 2025. [Online]. Available: https://api.semanticscholar.org/CorpusID:281585470

[56] H. Ren, M. Xia, L. Weng, H. Lin, J. Huang, and K. Hu, "Interactive and supervised dual-mode attention network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5612818.

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[57] S. Zhao, X. Zhang, P. Xiao, and G. He, "Exchanging dual-encoder–decoder: A new strategy for change detection with semantic guidance and spatial localization," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4508016.

[58] S. Dong, Y. Zhu, G. Chen, and X. Meng, "EfficientCD: A new strategy for change detection based with bi-temporal layers exchanged," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5639313.

[59] G. Pei and L. Zhang, "Feature hierarchical differentiation for remote sensing image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022. [Online]. Available: https://api.semanticscholar.org/CorpusID:251089099

[60] Z. Chen, H. Chen, J. Leng, X. Zhang, Q. Gao, and W. Dong, "VMMCD: VMamba-based multi-scale feature guiding fusion network for remote sensing change detection," *Remote Sens.*, vol. 17, no. 11, p. 1840, May 2025. [Online]. Available: https://api.semanticscholar.org/CorpusID:278920365

[61] Y. Feng, J. Jiang, H. Xu, and J. Zheng, "Change detection on remote sensing images using dual-branch multilevel intertemporal network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401015.

[62] C. Xu et al., "Progressive context-aware aggregation network combining multi-scale and multi-level dense reconstruction for building change detection," *Remote Sens.*, vol. 15, no. 8, p. 1958, Apr. 2023. [Online]. Available: https://api.semanticscholar.org/CorpusID:258034751

[63] M. Han, T. Xu, Q. Liu, X. Yang, J. Wang, and J. Kong, "HFIFNet: Hierarchical feature interaction network with multiscale fusion for change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 4318–4330, 2025. [Online]. Available: https://api.semanticscholar.org/CorpusID:275486271

[64] S. Fang, K. Li, and Z. Li, "Changer: Feature interaction is what you need for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5610111. [Online]. Available: https://api.semanticscholar.org/CorpusID:252367220

[65] S. Fu, S. Dong, and X. Meng, "Beyond cross-temporal difference: Style-aligned and fusion-difference learning for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5631415. [Online]. Available: https://api.semanticscholar.org/CorpusID:279703915

[66] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019. [Online]. Available: https://api.semanticscholar.org/CorpusID:57190346

[67] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604816. [Online]. Available: https://ieeexplore.ieee.org/document/9467555/

[68] S. Liu et al., "Network and dataset for multiscale remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 2851–2866, 2025. [Online]. Available: https://api.semanticscholar.org/CorpusID:275032911

[69] L. Shen et al., "S2Looking: A satellite side-looking dataset for building change detection," *Remote Sens.*, vol. 13, no. 24, p. 5094, Dec. 2021. [Online]. Available: https://api.semanticscholar.org/CorpusID:236134438

[70] P. Chen, D. Hong, Z. Chen, X. Yang, B. Li, and B. Zhang, "FCCDN: Feature constraint network for VHR image change detection," 2021, *arXiv:2105.10860*.

[71] R. Hang, S. Xu, P. Yuan, and Q. Liu, "AANet: An ambiguity-aware network for remote-sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5612911. [Online]. Available: https://api.semanticscholar.org/CorpusID:268204920

[72] J. Ma, J. Duan, X. Tang, X. Zhang, and L. Jiao, "EATDer: Edge-assisted adaptive transformer detector for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5602015. [Online]. Available: https://api.semanticscholar.org/CorpusID:266368343

[73] S. Liu et al., "Full-scale change detection network for remote sensing images based on deep feature fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5617113. [Online]. Available: https://api.semanticscholar.org/CorpusID:277397556

[74] H. Zhang, K. Chen, C. Liu, H. Chen, Z. Zou, and Z. Shi, "CDMamba: Incorporating local clues into mamba for remote sensing image binary change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4405016. [Online]. Available: https://api.semanticscholar.org/CorpusID:270285961

[75] Z. Wu, X. Ma, R. Lian, K. Zheng, and W. Zhang, "CDxLSTM: Boosting remote sensing change detection with extended long short-term memory," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025. [Online]. Available: https://api.semanticscholar.org/CorpusID:273969514

[76] H. Zhang, Y. Teng, H. Li, and Z. Wang, "STRobustNet: Efficient change detection via spatial–temporal robust representations in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5612215. [Online]. Available: https://api.semanticscholar.org/CorpusID:276298966

[77] Z. Zhang, L. Bao, S. Xiang, G. Xie, and R. Gao, "B2CNet: A progressive change boundary-to-center refinement network for multitemporal remote sensing images change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11322–11338, 2024. [Online]. Available: https://api.semanticscholar.org/CorpusID:270272635

[78] F. Zhou, X. Zhang, H. Shuai, R. Hang, S. Zhu, and T. Geng, "Dual-granularity feature alignment for change detection in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 4487–4497, 2025. [Online]. Available: https://api.semanticscholar.org/CorpusID:275376197

[79] Z. Xie et al., "FSG-net: Frequency-spatial synergistic gated network for high-resolution remote sensing change detection," 2025, *arXiv:2509.06482*.

[80] J. Yuan, L. Wang, and S. Cheng, "STransUNet: A Siamese TransUNet-based remote sensing image change detection network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9241–9253, 2022. [Online]. Available: https://api.semanticscholar.org/CorpusID:253333483

[81] Y. Luo et al., "MDENet: Multiscale difference information-guided dual-temporal enhancement network for remote sensing change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 27412–27429, 2025. [Online]. Available: https://api.semanticscholar.org/CorpusID:282359134

[82] Z. Liu et al., "Swin transformer V2: Scaling up capacity and resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11999–12009. [Online]. Available: https://api.semanticscholar.org/CorpusID:244346076

[83] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, New Orleans, LA, USA, Jun. 2022, pp. 11966–11976. [Online]. Available: https://api.semanticscholar.org/CorpusID:245837420

[84] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," 2019, *arXiv:1905.11946*.

[85] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," 2018, *arXiv:1807.06521*.

[86] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2017, pp. 7132–7141. [Online]. Available: https://api.semanticscholar.org/CorpusID:140309863

[87] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803. [Online]. Available: https://api.semanticscholar.org/CorpusID:4852647

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